

pocketful

- a software factory in BAND Desktop
- Dark Factory AI Hackathon | lablab.ai | solo entry
- A band of 5 coding agents that plans work, implements it, and checks its own results.

The factory shape

- The actual entry is the factory, not the app.
- architect - plans the work (only seat that designs)
- builder - the only seat that writes production code
- spec-warden - gates every deliverable against the track spec (executable conformance: 25 CONFORMS, 0 DEVIATES)
- race-hunter - finds concurrency and adversarial failure modes
- regression-guard - baselines the golden suite; nothing ships below baseline
- Mandates stay track-generic by construction (track-specific detail = DQ).

What it built: pocketful

- Zero-dependency wallet-and-payments JSON API (Node.js stdlib only, 0 dependencies).
- Wallet CRUD, transfers with SHA-256 idempotency keys, integer minor units, durable commit before mutation.
- Stages done: 1 (JSON API) + 3 (concurrency hardening: overdraft storms, key races, adversarial inputs) + 4 (domain extension: wallet statement + deposit lookup).
- Stage 2 (responsive UI, exact data-testid) gated on the official track spec.

How it checks its own work

- Executable spec-warden: `factory/conformance-check.js` - every spec claim is code, not prose.
- Golden suite: 14 tests green; race-hunter sweep: 8 tests green; stage-4 gate: 5 tests green.
- Clean-container gate: boots under `unshare -rn` with zero outbound network (verified).
- Bad-work recovery: the replay-status incident is documented in `FACTORY.md` - the factory caught, diagnosed, and fixed its own misfire.

Measured costs

- Inference: free tiers only (NVIDIA NIM cascade -> OpenRouter free -> Mistral -> Gemini).
- Total spend: \$0.00.
- Golden suite 14 tests ~7 s wall; race sweep 8 tests ~6 s; conformance probe 25 checks ~2 s.
- Full provenance ships with the submission: dispatch log, shot log, test runs.

Autonomy evidence (Agent Teamwork = 25%)

- In the submitted run, the dispatched task per stage was the only human input - no steering, no approvals, no reruns.
- Dispatch log: `factory/dispatch-log.md`.
- Review changed things: the replay-status fix traces to the spec-warden's review, not to a human.
- Blockers requiring the human: BAND account + API key (room export + room recording), official track spec (UI testids).

Status and next

- Done: stage 1 API, stage 3 concurrency hardening, stage 4 domain extension, repo + cover public, 4/7 video shots, this deck.
- Gated on parent browser work: BAND account + API key -> room export + room recording (DQ item); official track spec -> stage-2 UI.
- Then: final video assembly, submission on lablab.ai (user-confirmed only).
- The factory is the entry: hand the same five mandates a different problem and the same shape of work comes out.